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Roll#SU92-BSSEM-F25-041

ASSIGNMENT #03(PART 2)

PROGRAM#01

```
#include <iostream>
```

```
using namespace std;
```

```
int calculate2Low(int arr[], int size) {
```

```
    int low = INT_MAX, secondLow = INT_MAX;
```

```
    for (int i = 0; i < size; i++) {
```

```
        if (arr[i] < low) { secondLow = low; low = arr[i]; }
```

```
        else if (arr[i] < secondLow && arr[i] != low) secondLow = arr[i];
```

```
    }
```

```
    return secondLow;
```

```
}
```

```
int calculate2High(int arr[], int size) {
```

```
    int high = INT_MIN, secondHigh = INT_MIN;
```

```
    for (int i = 0; i < size; i++) {
```

```
        if (arr[i] > high) { secondHigh = high; high = arr[i]; }
```

```
        else if (arr[i] > secondHigh && arr[i] != high) secondHigh = arr[i];
```

```
    }
```

```

        return secondHigh;
    }

int main() {
    int n, arr[100];

    cin >> n;

    for (int i = 0; i < n; i++) cin >> arr[i];

    cout << "2nd Lowest: " << calculate2Low(arr, n) << endl;

    cout << "2nd Highest: " << calculate2High(arr, n) << endl;

    return 0;
}*/

```

Program#02

```

#include<iostream>

#include<iomanip>

using namespace std;

double calculateSum(int arr[],int size){

double sum=0;

for(int i=0;i<size;i++) sum+=arr[i];

return sum;

}

int main(){

    int n;

```

```

int arr[100];

    cin>>n;

for(int i=0;i<n;i++)

cin>>arr[i];

cout<<fixed<<setprecision(2)<<calculateSum(arr,n);

}

```

Program#03

```

#include<iostream>

#include<iomanip>

using namespace std;

double celciustoFahrenHeit(double c){

    return (c*9.0/5.0)+32;

}

int main(){

    double celcius;

    cin>>celcius;

    cout<<fixed<<setprecision(1)<<celciustoFahrenHeit(celcius);

}

```

Program#04

```

#include<iostream>

using namespace std;

bool leapyear(int year){

```

```

        if((year%400==0 )|| (year%4==0 && year%100 !=0))

return true;

return false;

    }

int main(){

    int year;

    cout<<"Enter a year"<<endl;

    cin>>year;

    cout<<"Leap year is"<<leapyear(year);

    return 0;

}

```

Program#05

```

#include<iostream>

using namespace std;

int vowelcounter(char str[]){

    int count=0;

    for(int i=0;str[i]!='\0';i++){

        char c=tolower(str[i]);

        if(c=='a' || c=='e' || c=='i' || c=='o' || c=='u')

            count++;

    }

    return count;

}

int main(){

```

```

char word[100];

cout<<"Enter a word"<<endl;

cin>>word;

cout<<"Total number of vowels are:"<<vowelcounter(word)<<endl;

}

```

Program#06

```

#include<iostream>

using namespace std;

void bubbleSort(char arr[],int size){

    for(int i=0;i<size-1;i++){

        for(int j=0;j<size-i-1;j++){

            if(arr[j]>arr[j+1]) swap(arr[j],arr[j+1]);

        }

    }

}

int main(){

    char arr[5];

    cout<<"Enter 5 Letters:";

    for(int i=0;i<5;i++){

        cin>>arr[i];

    }

    bubbleSort(arr,5);

    cout<<"Sorted Letters:";

    for(int i=0;i<5;i++){

        cout<<arr[i]; }

}

```